

HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY
HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION 2023

BIOLOGY PAPER 1

8:30 am – 11:00 am (2 hours 30 minutes)

This paper must be answered in English

GENERAL INSTRUCTIONS

- (1) There are **TWO** sections, A and B, in this Paper. You are advised to finish Section A in about 35 minutes.
- (2) Section A consists of multiple-choice questions in this question paper. Section B contains conventional questions printed separately in Question-Answer Book B.
- (3) Answers to Section A should be marked on the Multiple-choice Answer Sheet while answers to Section B should be written in the spaces provided in Question-Answer Book B. **The Answer Sheet for Section A and the Question-Answer Book B for Section B will be collected separately at the end of the examination.**

INSTRUCTIONS FOR SECTION A (MULTIPLE-CHOICE QUESTIONS)

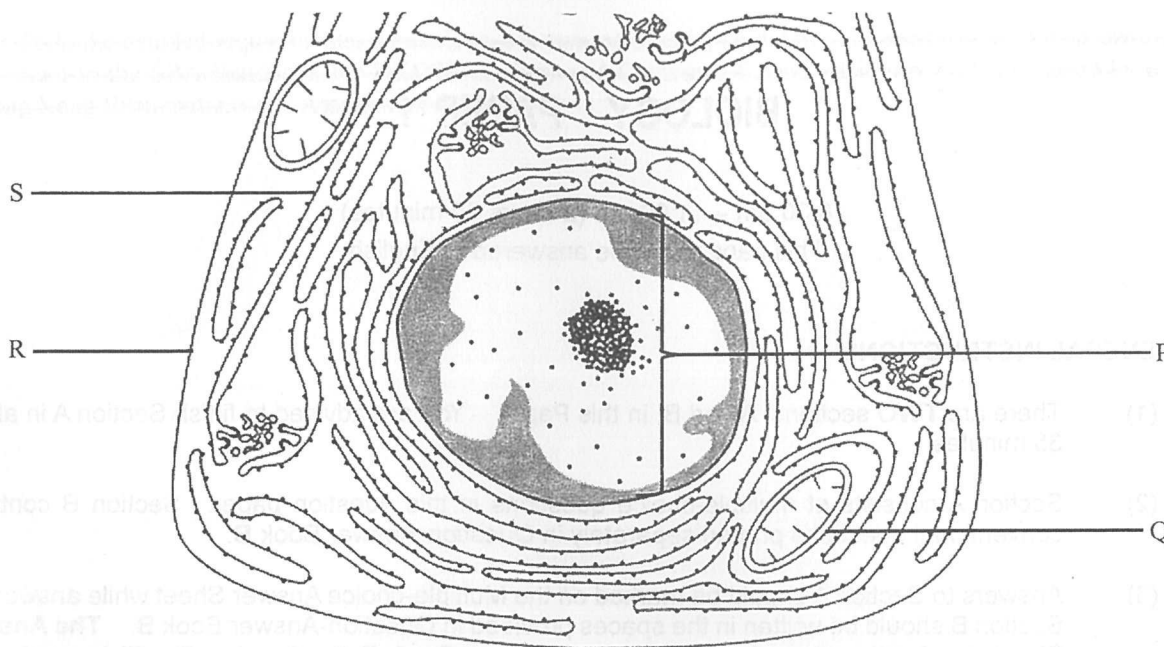
- (1) Read carefully the instructions on the Answer Sheet. After the announcement of the start of the examination, you should first stick a barcode label and insert the information required in the spaces provided. No extra time will be given for sticking on the barcode label after the 'Time is up' announcement.
- (2) When told to open this book, you should check that all the questions are there. Look for the words '**END OF SECTION A**' after the last question.
- (3) All questions carry equal marks.
- (4) **ANSWER ALL QUESTIONS.** You are advised to use an HB pencil to mark all the answers on the Answer Sheet, so that wrong marks can be completely erased with a clean rubber. You must mark the answers clearly; otherwise you will lose marks if the answers cannot be captured.
- (5) You should mark only **ONE** answer for each question. If you mark more than one answer, you will receive **NO MARKS** for that question.
- (6) No marks will be deducted for wrong answers.

Not to be taken away before the end of the examination session

There are 36 questions in this section.

The diagrams in this section are NOT necessarily drawn to scale.

Directions: Questions 1 to 3 refer to the schematic diagram below, which shows part of a pancreas cell:



1. Which of the following cellular components can also be found in a prokaryotic cell?

- A. P
- B. Q
- C. R
- D. S

2. If the pancreas cell is supplied with radioactively labelled amino acids, which of the following cellular components is likely to have an increase in radioactivity first?

- A. P
- B. Q
- C. R
- D. S

3. The radioactivity is later detected in the content of the small intestine. Which of the following substances is / are responsible for the presence of the radioactivity?

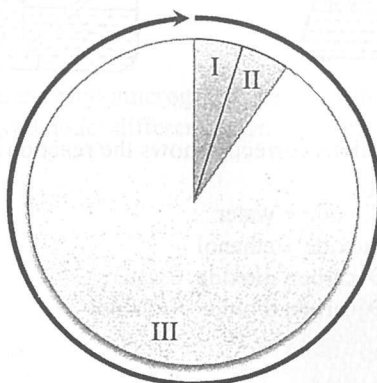
- (1) protease
- (2) glucagon
- (3) sodium hydrogencarbonate

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

4. Which of the following processes **does not** involve membrane proteins?

- A. transcription of DNA to form mRNA
- B. absorption of minerals by root hair cells
- C. digestion of carbohydrates in the small intestine
- D. transmission of nerve impulse between two neurones

Directions: Questions 5 and 6 refer to the diagram below, which shows three stages of a cell cycle:

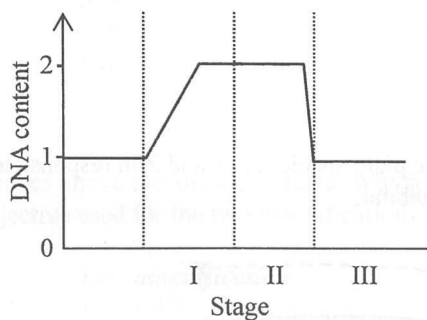


5. Which of the following correctly describes the events in the cell cycle?

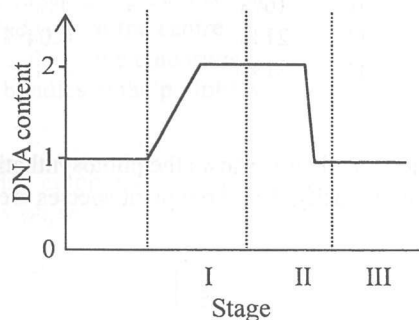
- A. The cytoplasm is halved in stage I.
- B. Sister chromatids separate in stage II.
- C. The cell synthesises more organelles in stage III.
- D. Chromatin condenses to form chromosomes in stage III.

6. Which of the following graphs correctly matches the changes in the DNA content in the cell with the three stages of the cell cycle?

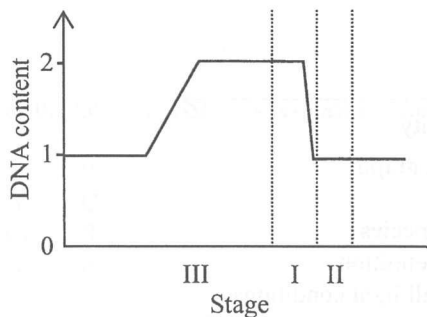
A.



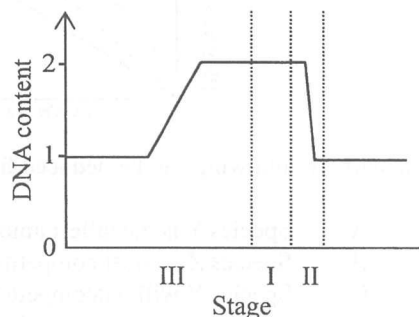
B.



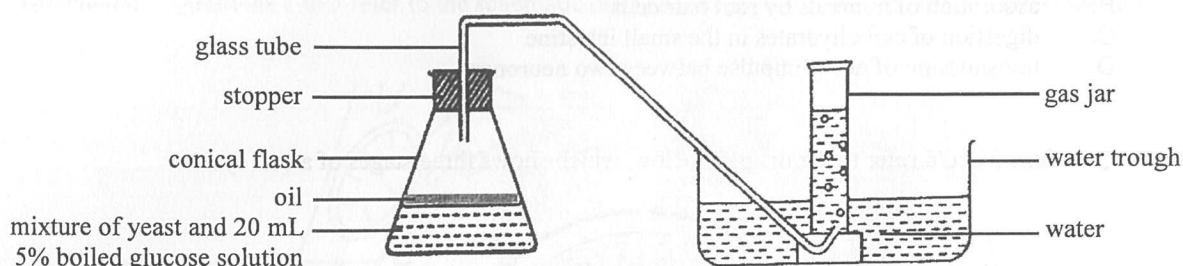
C.



D.



Directions: Questions 7 and 8 refer to the diagram below, which shows a set-up used to investigate the rate of respiration in yeast:



7. Which of the following word equations correctly shows the reaction being investigated?

- A. glucose $\rightarrow$ carbon dioxide + water
- B. glucose $\rightarrow$ carbon dioxide + ethanol
- C. glucose + oxygen $\rightarrow$ carbon dioxide + water
- D. glucose + oxygen $\rightarrow$ carbon dioxide + ethanol

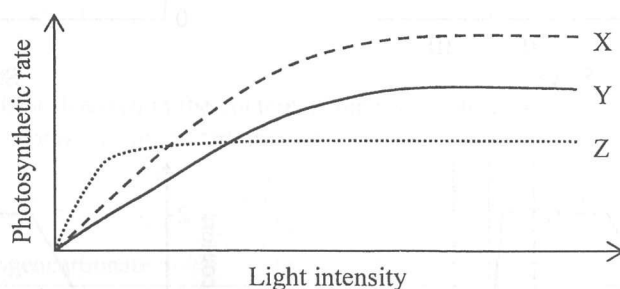
8. Which of the following modifications about the set-up can improve the accuracy of the experiment?

- A. use a measuring cylinder instead of a gas jar
- B. use a larger conical flask to contain the mixture
- C. use a glass tube with a smaller internal diameter
- D. use 30 mL instead of 20 mL of 5% boiled glucose solution

9. When we breathe out, some air will be left inside the air sacs in the lungs. Which of the following correctly shows the gas composition of the remaining air inside the air sacs?

	Oxygen	Carbon dioxide
A.	16%	0.04%
B.	16%	4%
C.	21%	0.04%
D.	21%	4%

10. The graph below shows the photosynthetic rates of three plant species X, Y and Z in response to the change of light intensity. The three plant species are in the same habitat.



Which of the following can be deduced from the above graph?

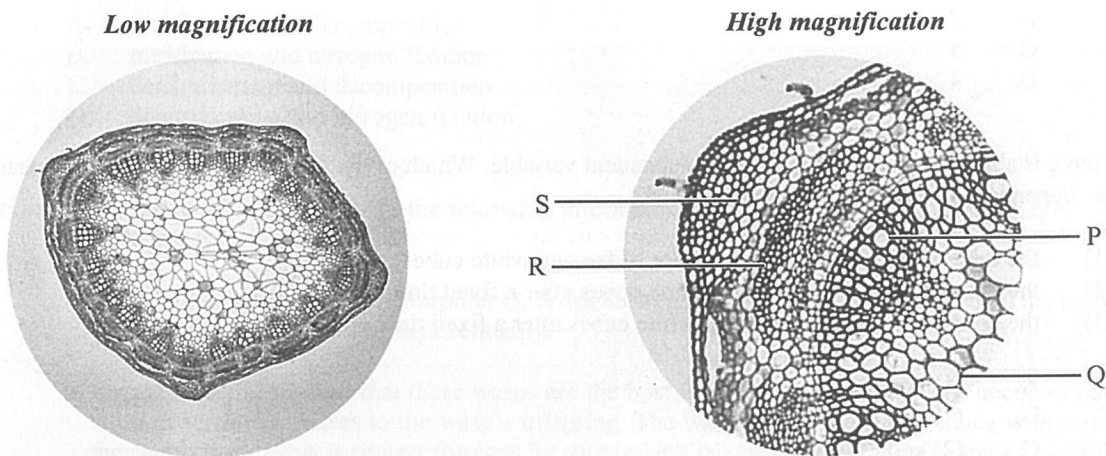
- A. Species Y is the tallest among the three species.
- B. Species Z is most competitive in shady condition.
- C. Species Y will outcompete species X in all light conditions.
- D. Removal of species Z will increase the photosynthetic rate of species X.

11. Which of the following processes produce ATP?

- (1) glycolysis
- (2) Calvin cycle
- (3) photochemical reactions

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Directions: Questions 12 to 14 refer to the photomicrographs below, which show the images of the cross section of a certain part of plant viewed under different magnifications:



12. Which of the following combinations correctly matches the part of the plant where the section was taken from and the supporting evidence from the photomicrograph?

	<i>Part of the plant</i>	<i>Evidence</i>
A.	root	presence of hairs on the epidermis
B.	root	with thin-walled cells at the centre
C.	stem	presence of cuticle on the epidermis
D.	stem	with vascular bundles at the periphery

13. The images above are shown to scale. Which of the following combinations correctly matches the eyepiece and objective used for the two magnifications (i.e. eyepiece x objective)?

	<i>Low magnification</i>	<i>High magnification</i>
A.	10X x 4X	10X x 10X
B.	10X x 4X	10X x 25X
C.	10X x 4X	10X x 40X
D.	10X x 10X	10X x 40X

14. Which of the labelled tissues is responsible for the transport of water?

- A. P
- B. Q
- C. R
- D. S

Directions: Questions 15 to 17 refer to an investigation about enzymes in fruits. Amy learned that some fruits contained proteases which can be used as meat tenderisers. She wanted to find a suitable fruit juice for the slow cooking of steak, i.e. mixing the steak with fruit juice in a sealed bag and then cooking it in a water bath set at around 50°C to 70°C for 1 hour. The procedure of her investigation is shown below:

1. Extract fruit juice from pineapple.
2. Add 15 mL of pineapple juice in 3 boiling tubes respectively with labels 50°C, 60°C and 70°C.
3. Put the tubes into water baths set at respective temperatures for 1 hour.
4. Let the tubes cool down to room temperature.
5. Add 3 egg white cubes of size 1 cm³ into each tube.
6. (Step for the measurement of the dependent variable)
7. Repeat the above steps with papaya, kiwi and lemon instead of pineapple.

15. Based on the procedure given, how many independent variable(s) is / are being studied?
 - A. 1
 - B. 2
 - C. 3
 - D. 4

16. Step 6 is about the measurement of the dependent variable. Which of the following can be used for measuring the dependent variable for this investigation?
 - (1) the time taken for the disappearance of the egg white cubes
 - (2) the change in shape of the egg white cubes after a fixed time
 - (3) the change in weight of the egg white cubes after a fixed time
 - A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

17. Which of the following is the assumption for this investigation?
 - A. The proteases are present in the fruits tested.
 - B. The proteases remain active at high temperatures.
 - C. The proteases are still functioning after extraction.
 - D. The actions of the proteases on egg white cubes and steaks are similar.

18. Which of the following features of the small intestine allow it to be the major site of food absorption?
 - (1) It is long.
 - (2) Its inner wall is highly folded.
 - (3) It secretes different kinds of enzymes.
 - A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

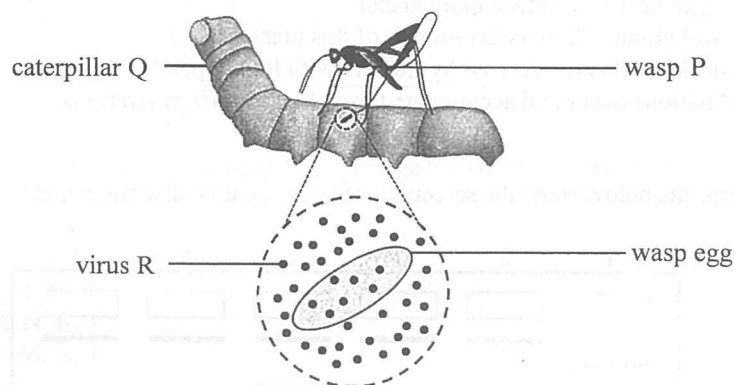
19. Blockage of the Eustachian tube will directly affect the functioning of the
 - A. cochlea.
 - B. eardrum.
 - C. ear bones.
 - D. oval window.

20. Which of the following correctly explain(s) why reflex actions are usually faster than voluntary actions?
- (1) Reflex actions do not involve decision making.
 - (2) Nerve impulses travel faster in reflex actions.
 - (3) Reflex actions are inborn.
- A. (1) only
 - B. (2) only
 - C. (1) and (3) only
 - D. (2) and (3) only
21. Kitchen wastes can be converted to fertilisers. Which of the following processes are essential for this conversion?
- A. nitrification and decomposition
 - B. nitrification and nitrogen fixation
 - C. denitrification and decomposition
 - D. denitrification and nitrogen fixation

Directions: Questions 22 and 23 refer to the following information about the interactions of wasp P, caterpillar Q and virus R.

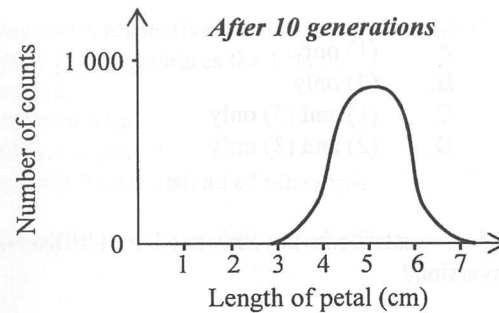
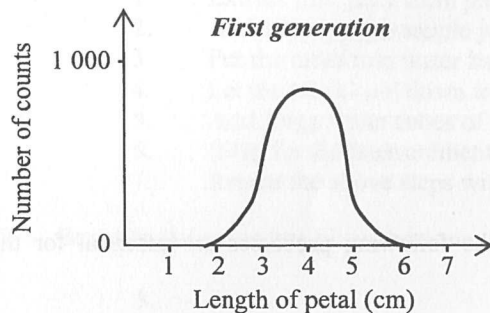
Wasp P lays eggs inside the caterpillar Q. When the eggs hatch, the wasp larvae feed on the body tissues of the caterpillars and kill them eventually.

Recent research showed that these wasps are the host of virus R. The viral DNA becomes part of the wasp genome and passes to the wasp's offspring. The wasps inject their eggs along with virus R into the caterpillars. Virus R protect the eggs by suppressing the caterpillars' body defence mechanism.

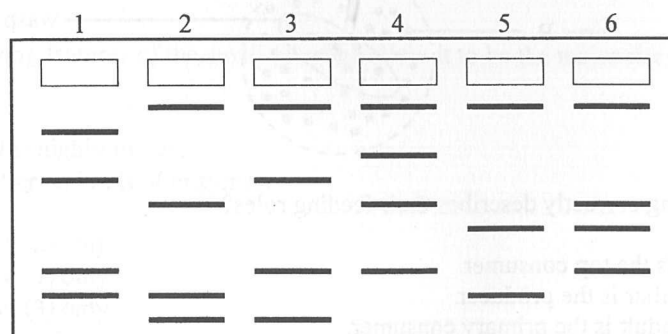


22. Which of the following correctly describes their feeding roles?
- A. The virus is the top consumer.
 - B. The caterpillar is the producer.
 - C. The wasp adult is the primary consumer.
 - D. The wasp larva is the secondary consumer.
23. Which of the following correctly shows the relationship between the wasp and the virus?
- A. parasitism
 - B. mutualism
 - C. competition
 - D. commensalism

Directions: Questions 24 and 25 refer to the information about the introduction of a foreign plant species to a local botanic garden. Over the years, the gardener noted that there was a change in the size of its flowers. The graphs below show the variations of the petal length of this plant species in the first generation and after 10 generations:



24. Which of the following correctly describe the inheritance of the petal length in this plant species?
- (1) The petal length is affected by environmental factors.
 - (2) The inheritance of the petal length is controlled by a number of genes.
 - (3) The variation in the petal length is a result of independent assortment.
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)
25. Which of the following is the most probable explanation for the change in the petal length of this plant species?
- A. Larger flowers produce more nectar.
 - B. Local climate favours the growth of this plant species.
 - C. Local insects are attracted by flowers with longer petals.
 - D. Mutations occur and accumulate throughout the 10 generations.
26. The DNA fingerprints below show the screening of a rare genetic disease in a family of six members:



The onset of the disease occurs in adulthood. Members 2 and 6 have already shown the symptoms of the disease. Which member is likely to have inherited this disease too?

- A. 1
- B. 3
- C. 4
- D. 5

Directions: Questions 27 and 28 refer to the following nucleotide sequence found on a mRNA molecule:

AUGGCUAACUAUAACGCU
 ↑
 translation starts

27. How many different codons are found on this nucleotide sequence?

- A. 3
- B. 4
- C. 5
- D. 6

28. Which of the following statements about the mRNA molecule are correct?

- (1) It can be found in the nucleus.
- (2) It can be found in the cytoplasm.
- (3) Its corresponding sequence on the DNA coding strand is TACCGATTGATATTGCGA.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Directions: Questions 29 and 30 refer to the phylogenetic relationship of five bacterial species. Enzyme X is found in most bacteria. The table below shows the number of nucleotide differences in the genes encoding enzyme X of five bacterial species (J, K, L, M and N):

	J	K	L	M	N
J	0	61	32	112	89
K	-	0	50	119	94
L	-	-	0	119	77
M	-	-	-	0	124
N	-	-	-	-	0

29. With regard to the phylogenetic relationship to L, arrange the species in the correct order from closely related to distantly related.

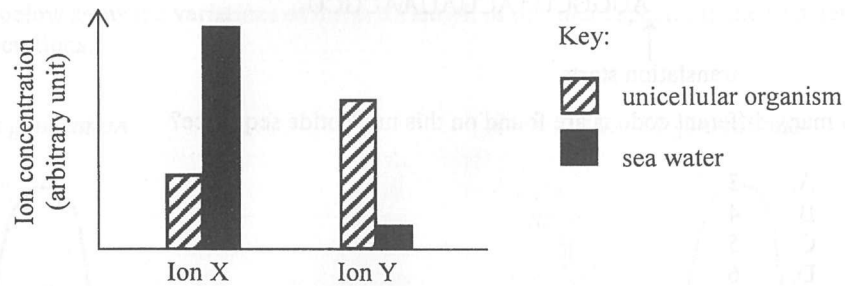
- A. L, J, K, N, M
- B. L, J, M, N, K
- C. L, M, N, K, J
- D. L, N, M, K, J

30. The nucleotide sequence of the genes encoding enzyme X have some differences among the five species. Yet, the enzyme X in these five species still serve the same function. Which of the following are possible explanations for this phenomenon?

- (1) The polypeptides formed have the same length.
- (2) There are more than one genetic code for the same amino acid.
- (3) The differences do not affect the shape of the active site of enzyme X.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

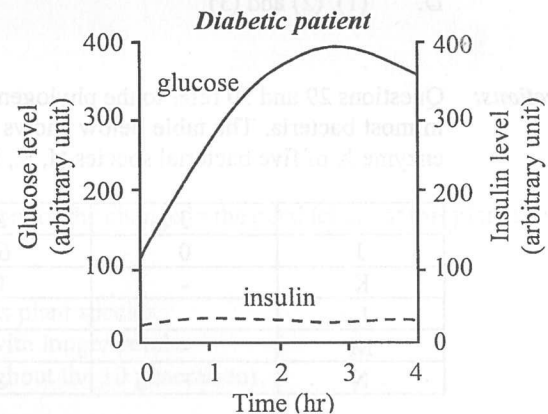
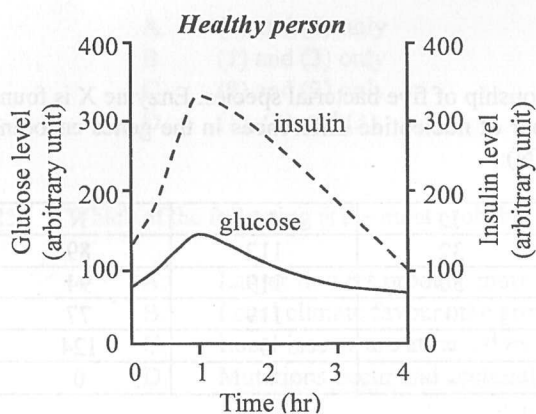
31. The graph below shows the concentration of ions in a unicellular organism and the sea water surrounding it:



Which of the following is most likely to occur in order to maintain the conditions shown in the above graph?

- Ion X moves out of the organism by diffusion.
- Ion X moves into the organism by active transport.
- Ion Y moves into the organism by active transport.
- Ion Y moves out of the organism by active transport.

Directions: Questions 32 and 33 refer to the graphs below, which show the changes in the blood glucose levels and blood insulin levels of a healthy person and a diabetic patient after consuming a sugary drink:



32. Based on the graphs, which of the following combinations correctly matches the type of diabetes the patient is suffering from and its explanation?

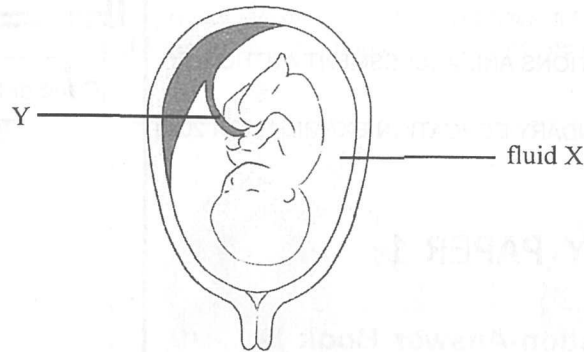
- | Diabetes | Explanation |
|-----------|---|
| A. type 1 | blood glucose level remains high after the patient consumes a sugary drink |
| B. type 1 | insulin level shows no response to the rise in blood glucose level |
| C. type 2 | blood glucose level has a sharp rise after the patient consuming a sugary drink |
| D. type 2 | liver fails to convert glucose to glycogen when blood glucose level is high |

33. The blood glucose level of the patient eventually drops because of

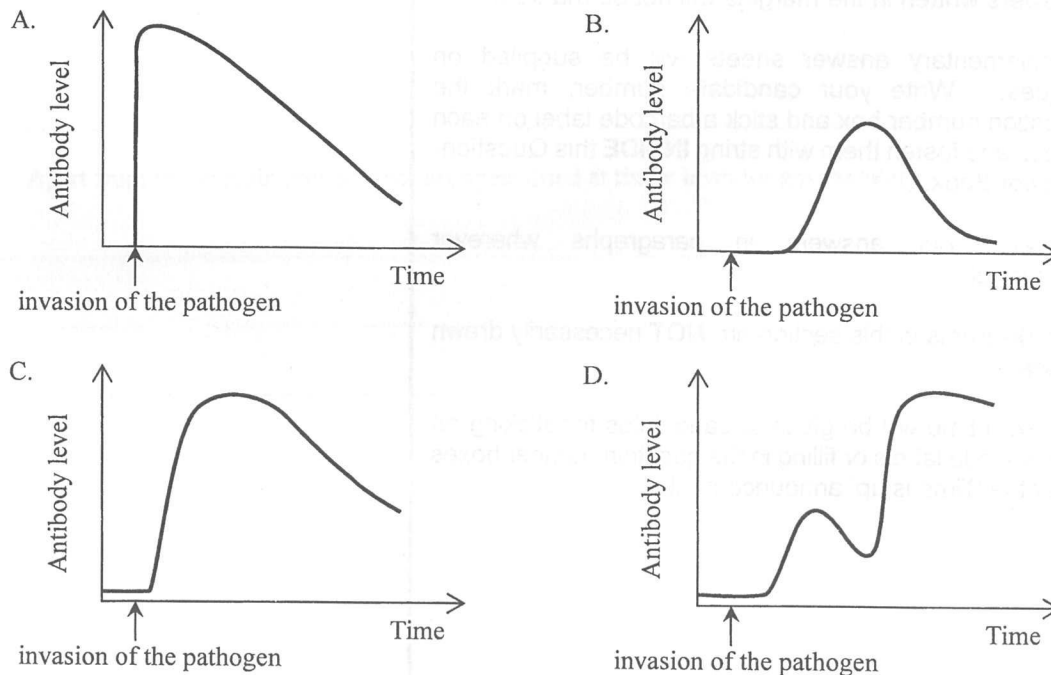
- the actions of glucagon.
- respiration of body cells.
- loss of glucose through urine.

- (3) only
- (1) and (2) only
- (2) and (3) only
- (1), (2) and (3)

Directions: Questions 34 and 35 refer to the diagram below, which shows a foetus and the associated structures inside the uterus:



34. Which of the following statements about fluid X is correct?
- It is secreted by the uterus.
 - It provides antibodies to the foetus.
 - It is a medium for material exchange.
 - It provides a stable environment for the foetus.
35. Which of the following combinations correctly compares the blood vessels inside structure Y?
- | | Artery inside Y | Vein inside Y |
|----|-----------------------------------|-------------------------------|
| A. | has a larger lumen | has a smaller lumen |
| B. | carries oxygenated blood | carries deoxygenated blood |
| C. | has less glucose in its blood | has more glucose in its blood |
| D. | blood flows towards foetus' heart | blood flows towards placenta |
36. A person is successfully vaccinated against infectious disease P. Which of the following graphs correctly shows the change in the antibody level of this person when he is invaded by the pathogen of disease P?



END OF SECTION A

Go on to Question-Answer Book B for questions on Section B